

PERSONAL INFORMATION



ABYAN ABDAN

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Nationality Indonesian

BIO

A graduate geologist with demonstrated experience in GIS and spatial data analysis applied to natural resource exploration and coal mining industry. I have worked in multidisciplinary roles and international environments, using geospatial data to support the evidence-based decision-making and the sustainable management of natural resources. Through the GEM MSc Geospatial Developer track, I seek to strengthen my technical skills and contribute to the innovation of geospatial solutions with global impact.

RELEVANT WORK
EXPERIENCE

(Dec 2023 – Dec 2025)

Supervisor Coal Quality Control

PT Manoor Bulatn Lestari (ESSAR Group)

West Kutai, East Kalimantan, Indonesia

An international company operating a multi-seam coal mining project.

- Supervised coal getting, hauling, and shipment, and conducted sample collection on each process.
- Performed daily assessment of coal quality, quantity, and temperature inspection. Managed database and produced report in the form of graphs and maps to support business decision-making.
- Initiated and developed standard operating procedures for coal laboratory operations, including sample preparation, proximate analysis, data analysis, data management, and consumable inventory management. Recorded more than 2000 coal analysis within a single year.
- Implemented rigorous quality control procedures which successfully increased coal production from 0.8 million Metric Tonnes in 2023 to 3.2 million MT in 2024 (+300% YoY) to 4.6 million MT in 2025 (+44% YoY) with zero product rejection.

(Mar 2022 – Dec 2023)

Coal Technologist/Geologist

PT SUCOFINDO (PERSERO) Tarakan Branch.

Nunukan, North Kalimantan, Indonesia

A state-owned company providing Testing, Inspection, and Certification (TIC) services.

- Led a quality control team responsible for coal quality and quantity monitoring from pit to port through systematic field sampling.
- Compiled and managed quality database providing data visualisations such as graphs and maps to improve reporting efficiency and enhance coal quality management.

(Apr 2019 – Sep 2020)

GIS-RS Analyst and Co-Founder

SPARVEON

Yogyakarta, Indonesia

A technology start-up specialising in geospatial solutions for natural resource exploration.

- Collected satellite imagery and geospatial datasets, performed spatial analysis, managed geodatabases, and produced final map with geological interpretations. Collaborated with the IT division to develop machine learning workflows for geoscience domain.

(Nov – Dec 2018)

GIS-RS Analyst

Satellite Imagery Processing and Spatial Modelling for Primary Tin Exploration in Nigeria, Africa.

A collaborative project between Universitas Gadjah Mada and PT Timah (PERSERO) Tbk.

- Analysed satellite imagery and regional geology data covering approximately 50 x 70 km² to identify new prospective areas for primary tin deposits.

(Feb – Jun 2017)

Geoinformation Laboratory Assistant

Laboratory of Mineral Resources, Geological Engineering Department, Universitas Gadjah Mada.

- Tutored second-year students in GIS software and geodatabase management for mapping projects.

EDUCATION

- (2014 - 2019) **S.T. (*Sarjana Teknik* or Bachelor of Engineering)**
Geological Engineering Department, Faculty of Engineering, Universitas Gadjah Mada, Indonesia
- (2011 - 2014) **SMA Yaspen Tugu Ibu 1 Depok**
Depok, Jawa Barat, Indonesia
- (2008 - 2011) **Escola Básica 2,3 Professor Delfim Santos**
São Domingos de Benfica, Lisbon, Portugal

PERSONAL SKILLS

Communication skills

- Structured communicator in Bahasa Indonesia and English for both technical and non-technical discussions.
- Active listener and reader in Portuguese for academic materials.
- Experienced in cross-cultural and multidisciplinary collaboration in geospatial research related to natural resources.

Organisational/managerial skills

- Collaborated with multidisciplinary team of nine professionals at SPARVEON (GIS, Remote Sensing, Geochemistry, Economic Geology, Geophysics, and Machine Learning).
- Led a team of 20 personnel at PT SUCOFINDO, managing daily operations, shift and roster leave scheduling, and troubleshooting.
- Coordinated with multiple departments at PT MBL Port Division involving more than 100 personnel.

Job-related skills

- End-to-end coal quality control procedures including field sampling, laboratory operations, temperature inspection, barge shipment, data analysis, and technical investigation.
- End-to-end GIS and remote sensing workflows for natural resource exploration including drone data acquisition, digital processing, field sampling, and data validation.

Digital competence

- GIS Desktop software (QGIS, ArcGIS Desktop, ArcGIS Pro, Global Mapper) – Intermediate, obtained during my bachelor studies and online courses.
- Remote sensing software (ENVI, ESA SNAP) – Intermediate, obtained during my studies to complete bachelor thesis project.
- Programming Language (Python, R, SQL, JavaScript) – Beginner, implemented the programming into geoscience and spatial analysis tasks since I joined SPARVEON team.
- Microsoft Office (Excel, Word, PowerPoint, Access, and PowerBI) – Intermediate, used for analytical reporting and managing tabular data in daily operational activities.

ADDITIONAL INFORMATION

Publications
Honours and awards

Academic Paper

"Identifikasi Kaolin Berdasarkan Analisis Citra ASTER di Daerah Kelabat, Kab. Bangka Barat, Provinsi Kepulauan Bangka Belitung" (2019). Authored by Abyan Abdan, S.T.; Dr. Agung Setianto, S.T., M.Si.; and Ir. Anastasia Dewi Titisari, M.T., Ph.D. Published in the Prosiding Seminar Nasional Kebumihan Geoweb ke-12.

Book

"Magmagma Expedition: The Secret of Bromo" (2019). Authored by Arif Muchlisin, et al. Published in the Gadjah Mada University Press.

Awards

- 1st Prize on Student Category. "ExploreSA: The Gawler Challenge" (2020), organised by Unearthed Solutions. Developed MineSAM, a machine learning system used for targeting various kinds of economic minerals in the Gawler Craton, South Australia.
- Finalist at "Airbus-Sobloo Multi-Data Challenge" (2019), organised by ESA Copernicus Masters. Proposed an idea called Xploradi, a remote sensing and AI based solution aiming to improve the exploration and prospecting of geothermal and mineral resources using big data processing.